

Bavin Saravanan

Portfolio: bavin-hub
Github: github.com/bavin-hub
Certificates

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EDUCATION

- **Carnegie Mellon University** Pittsburgh, United States
MS Mechanical Engineering ; GPA: 3.89/4.0 August 2025 - December 2026
Courses: Dynamics and Controls, Reinforcement Learning, Deep Learning, Artificial Intelligence, Machine Learning
- **SSN College of Engineering** Chennai, India
BE Mechanical Engineering ; GPA: 8.74/10.0 July 2019 - May 2023
Courses: Introduction to robotics, Mechatronics

SKILLS SUMMARY

Languages	Python, C++, JavaScript	Tools	Docker, Git, PostgreSQL, SQLite
Frameworks	PyTorch, TensorFlow, Keras, ROS2, Flask	Platforms	Linux, Web, Raspberry Pi, AWS, GCP

PUBLICATIONS

A Modular Simulation Suite and Benchmark for Aerial Manipulation Policy Learning

Under review, Conference on Robot Learning (CoRL), 2026

Yutong Wang, Dongjae Lee, Xiaofeng Guo, Yuanzhu Zhan, Yufei Jiang, **Bavin Saravanan**, Muqing Cao, Jia Xie, Chenyang Mao, Sebastian Scherer, Junyi Geng, Guanya Shi

Developed AM-Bench, a modular benchmark for aerial manipulation policy learning across 12 contact, transport, and constrained-interaction tasks. Evaluated how embodiment, low-level control, disturbances, and policy design interact under multirotor dynamics, revealing performance degradation and controller-policy coupling.

Comparing Surface Characteristics of Cu-Al-Fe Alloys Using Thermal-Based Machining Processes

Surface Engineering, vol. 40, no. 5, pp. 558–567, 2024 [view]

S. Santosh, **Bavin Saravanan**, T. S. Srivatsan

Compared EDM, Powder-Mixed EDM (PMEDM), and Laser Beam Machining on Cu-Al-Fe shape memory alloy using SEM, AFM, and roughness profiling; identified PMEDM as most effective, achieving minimum surface roughness of 2.377 μm .

EXPERIENCE

- **AirLab – CMU Robotics Institute** Onsite
Graduate Researcher Nov 2025 – Present
 - **Informative Path Planning:** Contributed to a multi-agent UAV system with class-based semantic exploration, enabling UAVs to prioritize regions by object categories, and designed a general swarm interface for scalable, coordinated autonomous exploration.
- **Legged Systems Group – CMU Robotics Institute** Onsite
Summer Research Intern Apr 2026 – Aug 2026
 - **RL Prosthesis Policy:** Contributed to the development and deployment of a reinforcement learning (RL) control policy for the CMU leg prosthesis, enabling online adaptation and stable, natural walking.
- **Skytex Unmanned Aerial Solutions** Onsite
AI and Robotics Engineer Jul 2023 – Jul 2025
 - **Decentralized Autonomous VTOL UAV Swarms:** Built a decentralized multi-agent VTOL system capable of autonomous navigation and collaborative mission execution in GPS-denied environments, with long-range and high-altitude flight capability.
 - **Autonomous UAV in Wild with AI Navigation and Perception:** Developed compact UAVs with structured search, real-time 3D mapping, and AI-based perception for autonomous navigation, obstacle avoidance, detection, and tracking in GPS-denied environments.

PROJECTS

- **Generalizable Decoupled Loco-Man:** Developed a reinforcement learning system to train the Unitree G1 humanoid for navigation and manipulation tasks using Isaac Lab 2.3+. The goal is to enable general locomotion and manipulation capabilities across different robotic embodiments. (Sept–Dec '25)
- **SimpleRL:** Implemented reinforcement learning algorithms from scratch for discrete and continuous environments based on research papers, focusing on core concepts rather than optimized performance. The project is designed for clarity and experimentation, supporting learning and further exploration of RL methods. (Dec '25)
- **Search Engine:** Built a RAG-based search engine with a React frontend and Python Flask backend, connected through REST APIs for efficient communication. The decoupled architecture allows flexible integration with any compatible backend system. (Jan '26–Present)
- **Jaffa-LLM:** Trained a native Tamil language model on multilingual C4, news articles, books, and Wikipedia using two A6000 GPUs with data parallelism. It was further fine-tuned on conversational Tamil data to enable coherent multi-turn dialogue. (Jul–Aug '24)